

String Manipulation

In this question, you will be implementing the following functions.

```
int findChar(char * str, char c);
```

Searches for the character `c` in the string `str` and returns the index of the character in the string. If the character does not exist, returns -1

```
int replaceChar(char * str, char c1, char c2);
```

Searches for the character `c1` in the string `str` and if found, replace it with `c2`. The function returns the number of replacements it has performed. If the character does not exist, returns 0.

```
int removeChar(char * str1, char * str2, char c);
```

Creates a copy of `str1` into `str2` except for the character `c` that should be replaced with `'*'`

For example, if

`str1="Hello World"` and

`c='l'` then the function should make

`str2="He**o Wor*d"`

```
int isPalindrome(char * str)
```

Checks to see if a string is Palindrome(reversible). If it is, returns 1, otherwise returns 0. A palindrome string reads similarly from left to right and from right to left like madam, level, radar, etc.

```
int reverseString(char * str)
```

Gets a string, reverses it and returns the number of characters in that string.

Develop a C code that implements the functions described above and use main function to call and test your code.